

Stress Detection via Multimodal Multitemporal-Scale Fusion: A Hybrid of Deep Learning and Handcrafted Feature Approach

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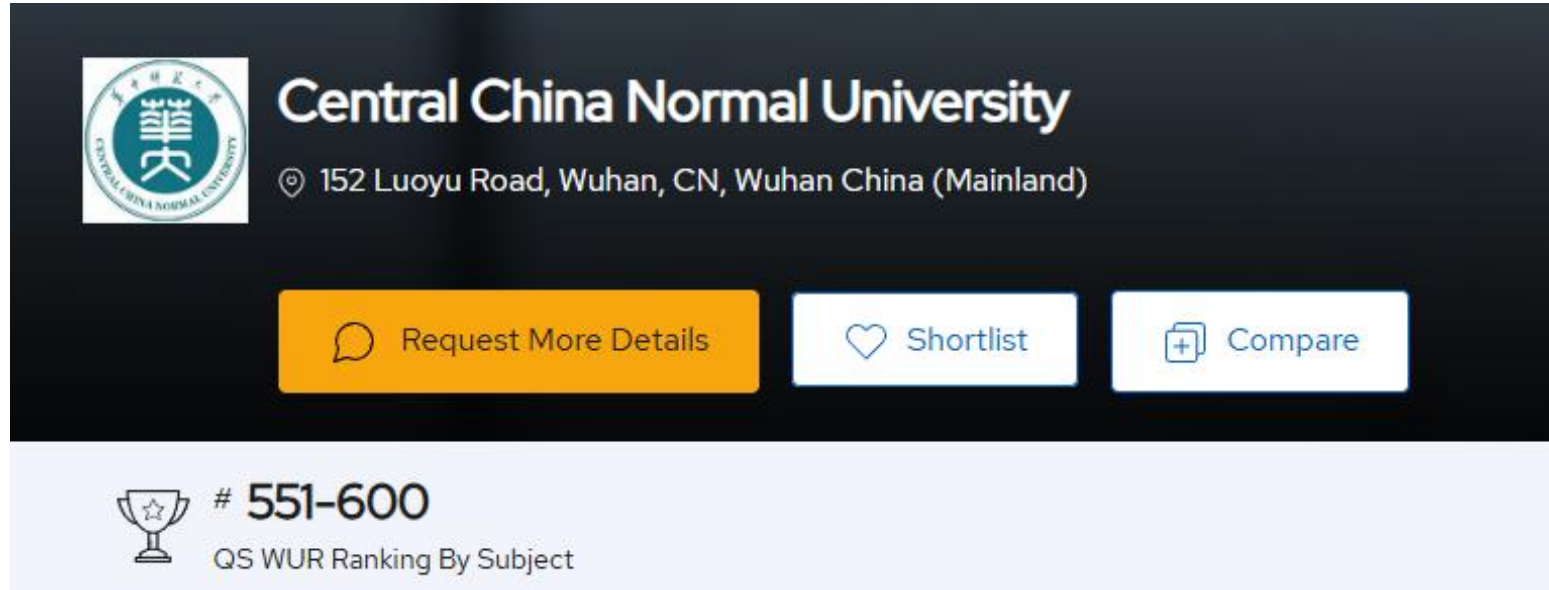
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ENGINEERING, ELECTRICAL & ELECTRONIC

84/353

JCR YEAR	JIF RANK	JIF QUARTILE	JIF PERCENTILE
2023	84/353	Q1	76.3

Rank by JIF before 2023 for ENGINEERING, ELECTRICAL & ELECTRONIC

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Science Citation Index Expanded (SCIE)

JCR YEAR	JIF RANK	JIF QUARTILE	JIF PERCENTILE	
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2021	83/276	Q2	70.11	
2020	96/273	Q2	65.02	
2019	91/266	Q2	65.98	

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JCR Category	Category Rank	Category Quartile
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OUTLINE

- ◆ Introduction
- ◆ Methodology
- ◆ Results
- ◆ Conclusions

Motivation

- Stress is common in modern life, affecting health, behavior, and performance, making detection and management essential.
- Stress disrupts autonomic balance, causing changes in heart rate, blood pressure, and respiration. Multi-source signals offer an efficient way to detect stress.



Research Gap

Authors	Data Source	Algorithm	Feature selection	Accuracy
Schmidt et al. [37], 2018	All physio	AB	Handcrafted	79.86%
Sah et al. [68], 2022	All signals	SVC	Handcrafted	81.35%
Lai et al. [69], 2021	All physio	Res-TCN	DL	86.50%
Martin et al. [70], 2022	All physio	CNN	DL	85.10%
Proposed Method	ECG+EDA	XGB-Boost	DL+Handcrafted	90.05%



Contributions

- A stress detection system is proposed based on ECG and EDA with multiple temporal scales to take advantage of multisource physiological signals.
- A hybrid approach of handcrafted and deep learning (DL) features is implemented to gain high accuracy.
- A multimodal fusion-based stress classification is proposed to construct a generalizable detection system.

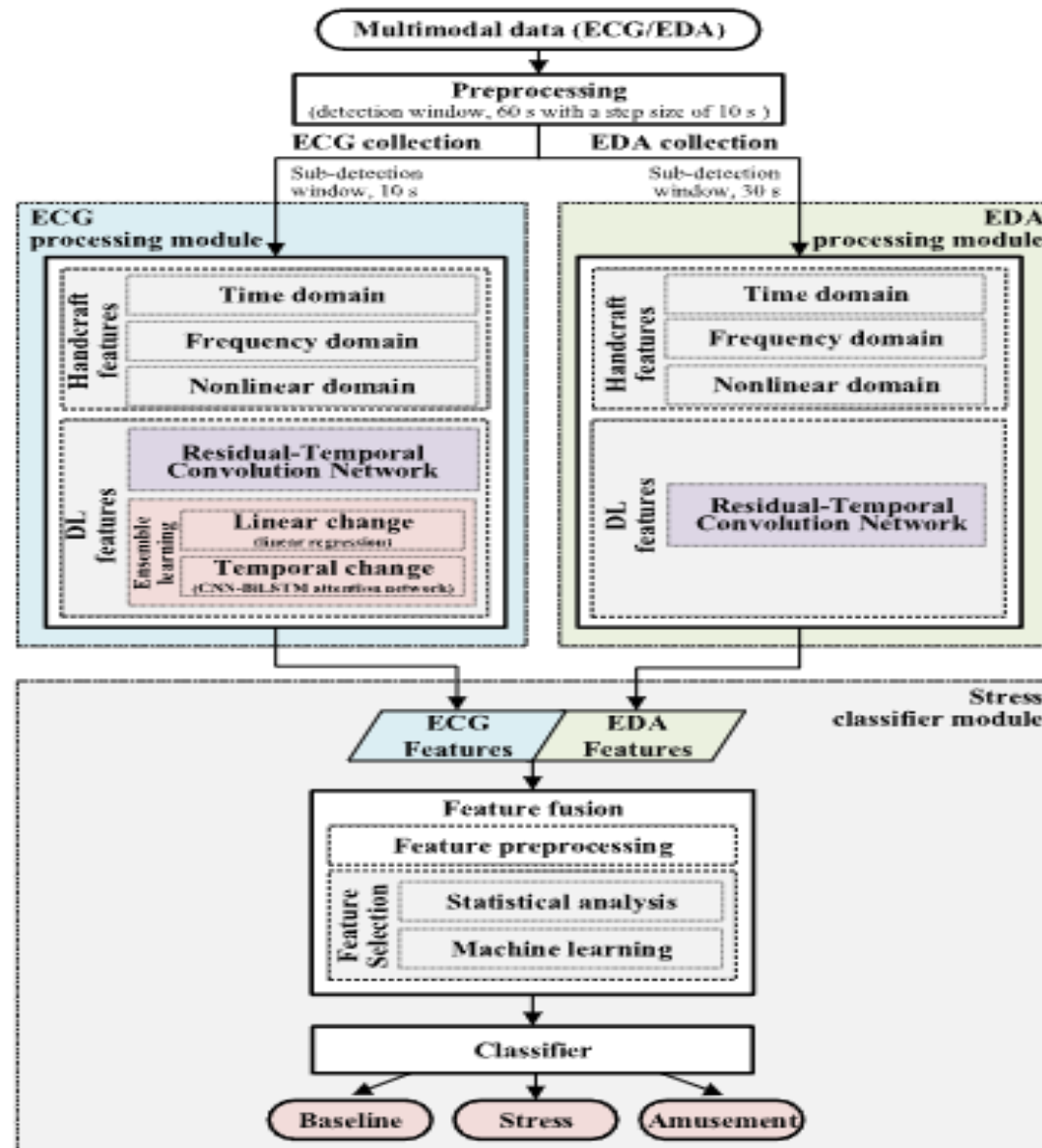


Dataset

- The WESAD dataset contains multiple types of data collected from 15 subjects, including ECG, EDA, respiration, accelerometer, skin temperature, and EMG.
- Data collection covers three different conditions: baseline, amusement, and stress, where stress is induced through public speaking and mental arithmetic tasks.
- The study used RespiBAN and Empatica E4 devices, with an average age of 27.5 years for the subjects, consisting of 12 males and 3 females.



Flowchart

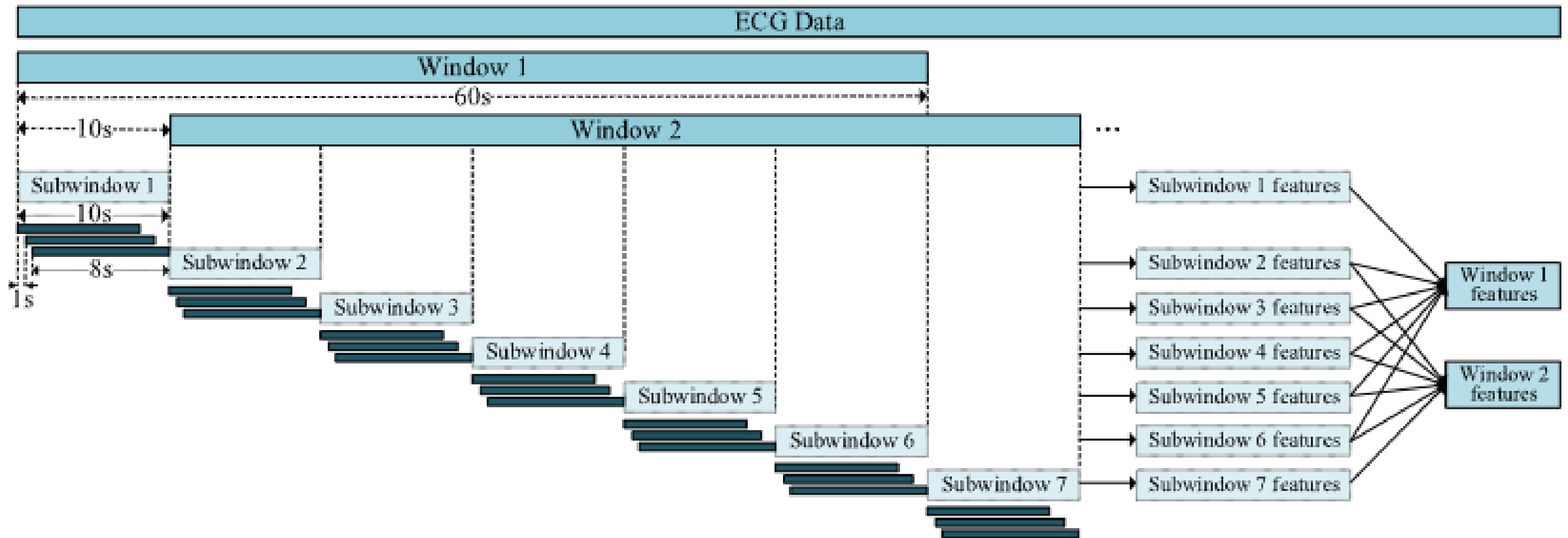


ECG Processing Module

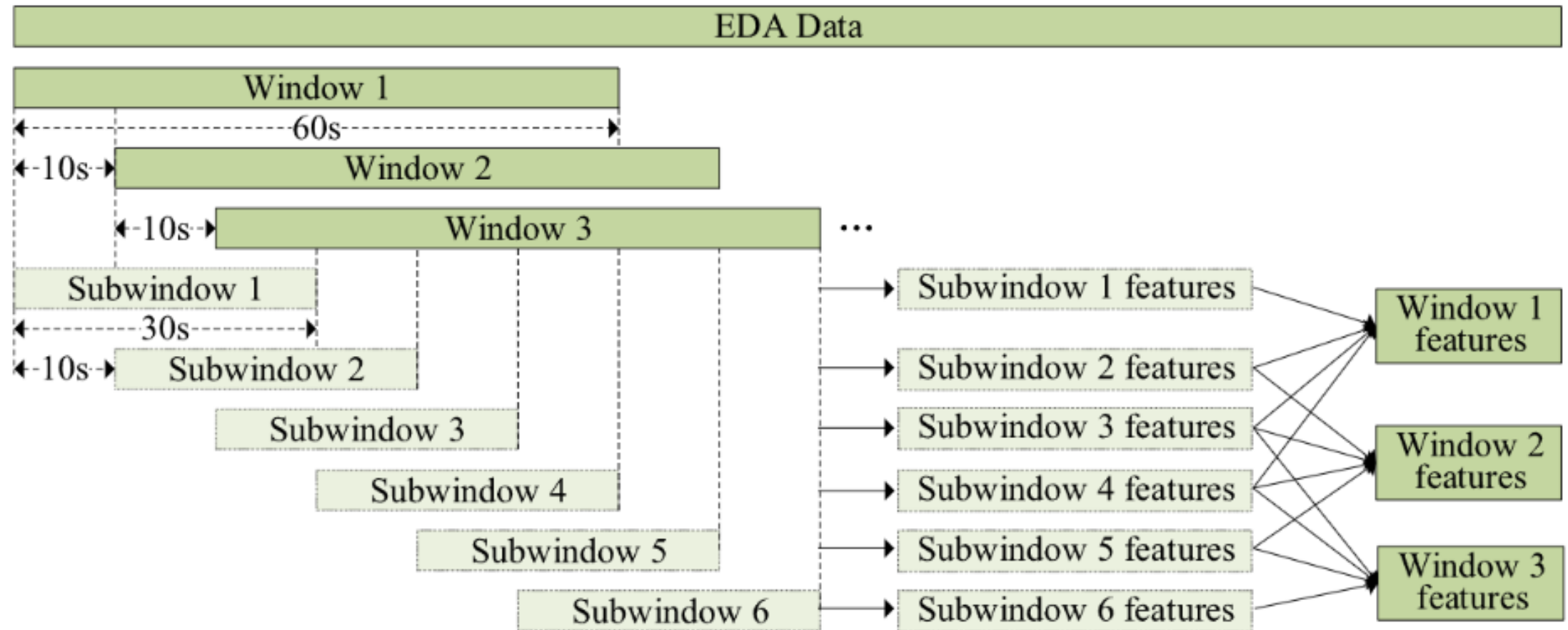
- R-R intervals are calculated by detecting R peaks, and HR , HRV and other features, including time-domain, frequency-domain, and nonlinear-domain features such as pnn20 and Baevsky's Stress Index (SI), are extracted.
- Features are extracted across both the detection window (60 seconds) and the sub-detection window (10 seconds) to achieve more precise stress pattern recognition.



Sliding Window For ECG Data Processing



Sliding Window For EDA Data Processing



Handcrafted Features

- EDA
 - **Time domain:** ① SC features: mean, std, median, mode, mean of derivative, mean of second derivative, dynamic range, slope, maximum, minimum, sum, skewness, kurtosis, lag, mean period, difference between maximum and minimum; ② SR features: mean, std, mean of derivative, mean of negative derivative, proportion of negatives in derivative, number of local minimum, proportion of local minimum, mean of each rise time series, TPR, MTPR; ③ SCR features: mode, sum of startle magnitudes, sum of response durations, area under the identified SCRs, mean, std, sum amplitude, number of SCR peaks, slope, dynamic range, skewness, kurtosis, sum of peak prominences, sum of peak widths, maximum, sum of peaks, 20th percentile, 80th percentile, root mean square of peaks; ④ SCL features: mean, std, correlation between SCL and time; ⑤ SCSR features: mean, std, mean of derivative, mean of second derivative; ⑥ SCSR & SCVSR features: zero crossing rate of SCSR, zero crossing rate of SCVSR, average magnitude of SCSR occurrences, average magnitude of SCVSR occurrences, average magnitude of all SCSR and SCVSR occurrences, ratio of number of SCSR occurrences to SCVSR
 - **Frequency domain:** ① SC features: mean, std, sum, maximum, minimum, dynamic range, difference between maximum and minimum; ② SR features: spectral power 0 – 2.4 Hz (12, per 0.2 Hz)
 - **Nonlinear domain:** ① SC features: sample entropy, LyE, HurstE, DFA, RCMDE (40); ② SR features: Shannon entropy, RCMSE (20), RCMPE (100); ③ SCR features: Shannon entropy of peaks

Data	Extracted Features
● ECG	○ Time domain: ① HR & HRV features: <u>mean</u>, <u>std</u> (<i>standard deviation</i>), <u>median</u>, <u>mode</u>, <u>rms</u> (<i>root mean square</i>), <u>sdnn</u>, <u>rmssd</u>, <u>pnn20</u>, <u>nn20</u>, <u>pnn50</u>, <u>nn50</u>, <u>sd1</u>, <u>sd2</u>, <u>sd1/sd2</u>, <u>sdsd</u>, skewness, kurtosis, <u>percentage of points higher than mean + std</u>, <u>percentage of points lower than mean – std</u>, TINN, SI; ② <u>ppg45 features</u> (39); ③ Waveform-related features: <u>TPR</u>, <u>MTPR</u>, <u>lag</u>, mean period, <u>svri</u> ○ Frequency domain: ① Spectral power/ratios: <u>0 – 6 Hz</u> (60, <u>per 0.1 Hz</u>), LF, MF, HF, <u>TF</u>, <u>peakLF</u>, <u>peakHF</u>, peakLF/peakHF, <u>LF/HF</u>, <u>LF/TF</u>, HF/TF, <u>LF/(TF-LF)</u>, and HF/(TF-LF); ② <u>ppg45 features</u> (6) ○ Nonlinear domain: <u>RCMSE</u> (12), <u>RCMPE</u> (15), <u>RCMDE</u> (15), <u>Shannon entropy</u>, <u>sample entropy</u>, <u>LyE</u>, <u>HurstE</u>, <u>DFA</u>

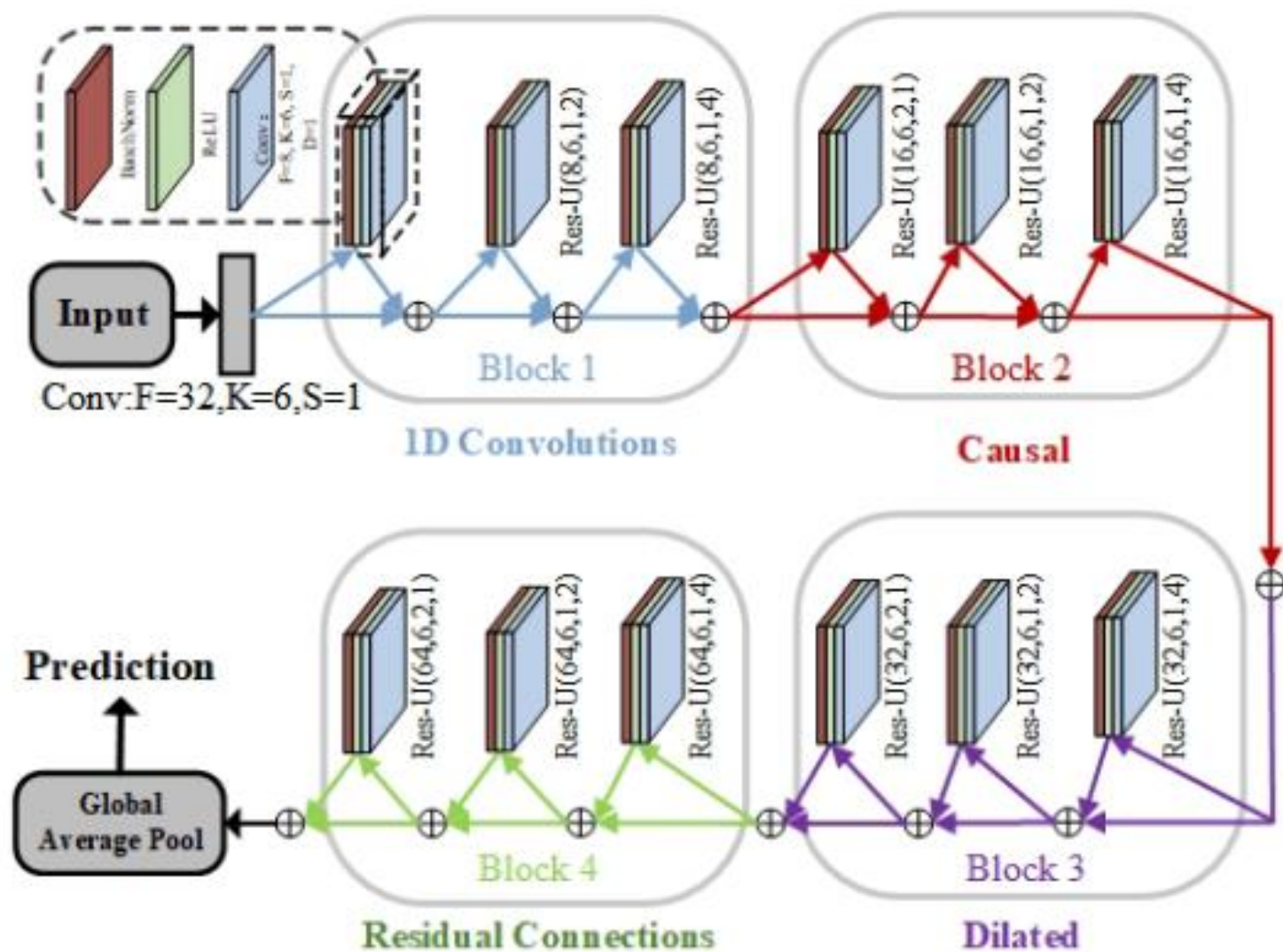


DL Features

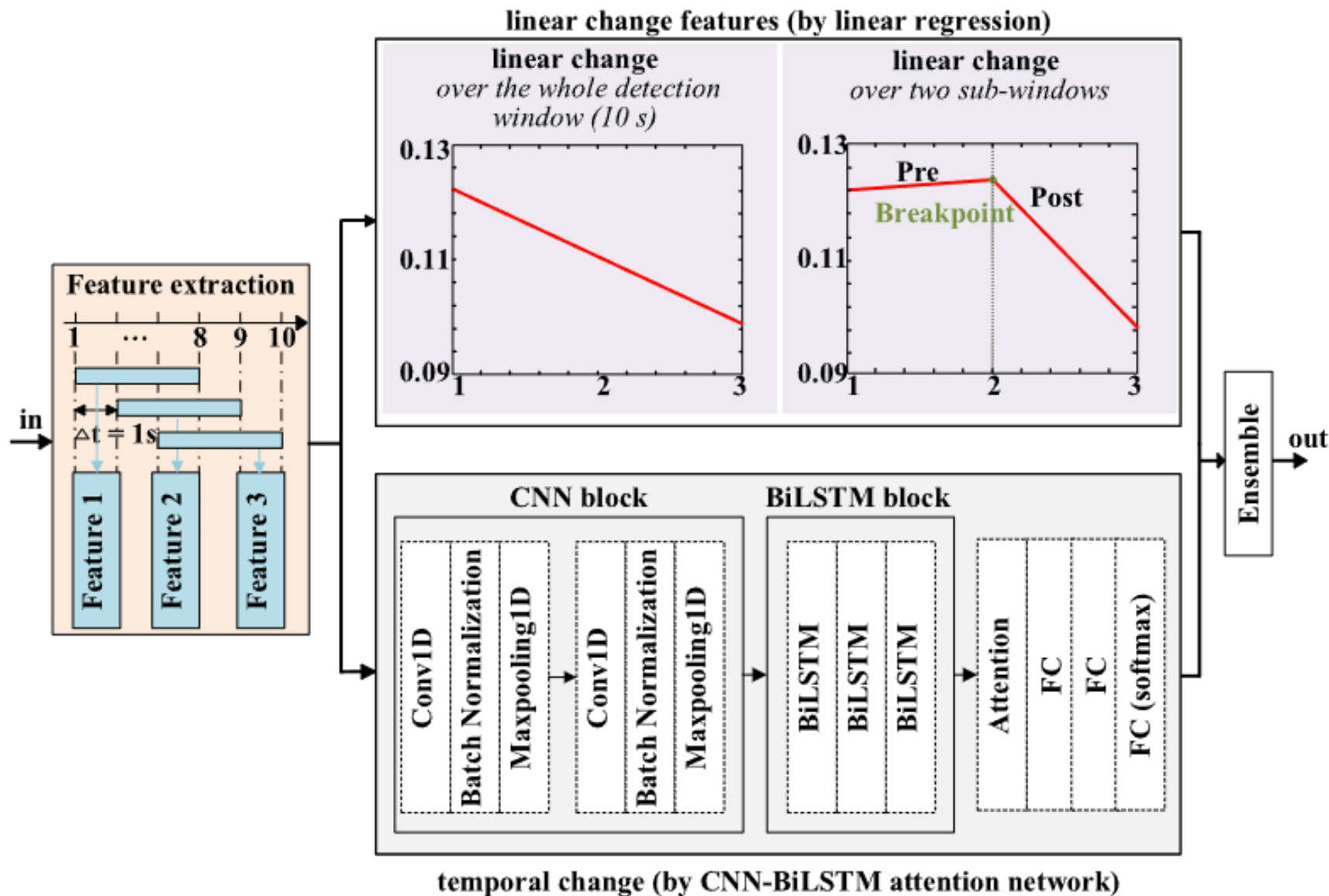
- In the 60-second detection window, a Res-TCN is utilized to extract deep learning features. It enhances the model's ability to learn time-series data through 1-D convolution, causal profilers, dilated convolution, and residual connections, accelerating training convergence.
- In the 10-second subdetection window, an ensemble learning model is applied, combining handcrafted features and deep learning models (e.g., CNN-BiLSTM) to extract features from smaller windows. The ensemble learning approach improves prediction accuracy.



Res-TCN



Ensemble Learning Model



EDA Processing Module

- Time-domain, frequency-domain, and nonlinear-domain features are extracted from different EDA signals, such as skin conductance, slow responses, and rapid responses.
- Motion artifacts are removed using a 3 Hz low-pass filter, and the EDA data is separated into slow-changing components (SCL) and rapidly changing components (SCR).
- Res-TCN is used once more for extracting DL features from EDA data.



Classification Module

- ECG and EDA features from different time windows are aggregated, followed by preprocessing, including filling missing values and individual normalization.
- Recursive Feature Elimination (RFE), Random Forest, and SelectKBest are used to select 46 features strongly related to stress status. PCA is then performed to reduce dimensions, generating two new features.
- The selected 48 features are input into an Extreme Gradient Boosting (XGBoost) classifier.



Experimental Results

Method	Data Sources	No. of wearables	Accuracy (%)	F1 Score (%)	
[37]	ECG	1	66.29	56.03	
	EDA	1	67.07	46.83	
	All physio	8	79.86	71.10	
	All signals	10	79.57	68.85	
[63]	EDA	1	66.12	74.19	
[64]	ECG	1	69.87	65.91	
	EDA	1	74.44	70.78	
	ECG+EDA	2	80.18	76.38	
[65]	ECG	1	75.35	80.03	
[67]	EDA	1	67.23	70.49	
[68]	ECG	1	76.27	--	
	EDA	1	70.03	--	
	All signals	10	81.35	--	
[69]	ECG	1	72.66	68.36	
	EDA	1	68.58	64.14	
	All signals	10	86.50	83.01	
[70]	All physio	8	85.10	85.05	
	All signals	10	85.03	85.01	
Proposed	Hand-crafted	ECG	1	75.21	67.87
		EDA	1	74.12	70.08
		ECG+EDA	2	75.22	72.36
	DL	ECG	1	86.23	82.84
		EDA	1	65.57	73.75
		ECG+EDA	2	86.40	83.00
	Hand-crafted	ECG/wo	1	76.53	72.65
		EDA/wo	1	79.93	76.82
		ECG+EDA/wo	2	83.19	80.39
	+ DL	ECG	1	85.83	82.67
		EDA	1	81.85	79.57
		ECG+EDA	2	90.05	86.78

— handcrafted
— DL



Conclusions

- This study proposes a stress detection system based on ECG and EDA data, leveraging multisource physiological signals to improve detection accuracy.
- A multi-scale feature fusion method combines handcrafted and deep learning features to improve detection accuracy.
- The proposed system improves detection accuracy by 9.56% and 3.55% compared to the best single-method ECG and EDA-based approaches.



Comments

- Pros

- The processing steps for each signal are explained in detail, and the study also replicates other related works to ensure the confidence in the comparisons.

- Cons

- This study only used one dataset to validate the method, and the complex processing steps may increase the computation time, thereby raising the system's time complexity.

- Future direction

- The feasibility of this method in real-world environments needs to be validated, and currently, chest-worn sensors are used. In real life, more discreet and comfortable sensors would be required.



Open code / Open data

- Code not available.
- Datasets available.

WESAD:

https://ubi29.informatik.uni-siegen.de/usi/data_wesad.html



THANK YOU!

